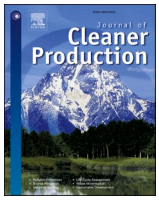
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Large language models for life cycle assessments: Opportunities, challenges, and risks

Nathan Preuss [a], Abdulelah S. Alshehri [b,c], Fengqi You [a,b,d,*]

[a] Systems Engineering, Cornell University, Ithaca, NY, 14853, USA
[b] Robert Frederick Smith School of Chemical and Biomolecular Engineering, Cornell University, Ithaca, NY, 14853, USA
[c] Department of Chemical Engineering, College of Engineering, King Saud University, Riyadh, 11421, Saudi Arabia
[d] Cornell Atkinson Center for Sustainability, Cornell University, Ithaca, NY, 14853, USA

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ABSTRACT

Because sustainability remains a wicked problem, more sophisticated tools need to be applied to identify better solutions in a more efficient manner and align with the 11th, 12th, and 13th sustainable development goals: sustainable cities and communities, responsible consumption and production, and climate action. To ease the burdens of conducting sustainability studies, especially life cycle assessments (LCA), practitioners may consider integrating large language models (LLM) into LCAs. This emerging application may offer some advantages due to the capability of these models to generate and process text quickly and efficiently, decreasing the time it takes to complete an LCA and increasing the accessibility of LCAs. In this perspective, we assess the ability of LLMs to complete LCA tasks and encourage the LCA community to study the potential strategies for enhancing the integration of LLMs in LCA methodologies and collaborate to develop standards for responsible use. Because of these advantages, LLMs show promise for life cycle inventory data collection and interpreting the life cycle impact assessment. Challenges arise primarily from the inclusion of hallucinations in the content generated by the LLM, which can be mitigated if the LCA practitioner uses prompt engineering techniques. Moreover, the risk that models cannot take responsibility for generated content can be ameliorated by having the LCA practitioner carefully review the LLM output and take responsibility for decisions made based on the generated content. So long as appropriate steps are taken to overcome the challenges and risks of using of LLMs for LCA, the opportunities presented by integrating the generative AI models can streamline the LCA process and result in significant benefits for the LCA practitioner.

1. Introduction

Sustainability challenges are wicked problems; there is no definitive formulation of the problem and no "correct" answer (Lönngren and van Poeck, 2021). Sustainability challenges impact a variety of sustainable development goals (SDG) (United Nations); for example, avoiding water scarcity (He et al., 2021) and ensuring access to clean water (Mueller and Gasteyer, 2021) as part of SDG 6, protecting forests (Buchadas et al., 2022) and preserving biodiversity (Ly et al., 2023) as part of SDG 14 and 15, and providing clean energy (Dincer and Acar, 2015) as part of SDG 7. To evaluate potential solutions to these problems and encourage sustainable cities, responsible consumption and production, and climate action as part of SDGs 11, 12 and 13, environmental assessments are often conducted (Pope et al., 2004). While there are various types of environmental assessments (Loiseau et al., 2012), life cycle assessments (LCA) (Hauschild et al., 2018) are a widely-used type of sustainability study that has standardized tasks (Finkbeiner et al., 2006). When LCAs are conducted by humans, there are numerous opportunities for error that may distort the results given the difficult nature of the problems being assessed (Reap et al., 2008a), (Reap et al., 2008b). Moreover, LCAs are time intensive, often taking months to complete. Therefore, there are numerous opportunities to improve the quality of the LCA while simultaneously decreasing the time necessary to complete an accurate LCA. To address these problems, large language models (LLMs) can be utilized to manipulate, generate, and disseminate text in a quick and efficient manner and can even surpass human-level understanding and processing capabilities (Gemini Team et al., 2023). LLMs are large artificial intelligence (AI) models that act as sophisticated tools to mimic human writing (Naveed et al., 2023). When integrated with multi-modal

* Corresponding author. Systems Engineering, Cornell University, Ithaca, NY, 14853, USA.
E-mail address: fengqi.you@cornell.edu (F. You).

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0959-6526/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Abbreviations

AI	Artificial intelligence
CoT	Chain of thought
ESG	Environmental, social, governance
LCA	Life cycle assessment
LCI	Life cycle inventory
LCIA	Life cycle impact assessment
LCO	Life cycle optimization
LLM	Large language model
ML	Machine learning
MM-LLM	Multi-modal large language model
PCF	Product carbon footprint
RAG	Retrieval augmented generation
RLHF	Reinforcement learning with human feedback
SDG	Sustainable development goal
VDB	Vector databases

LLMs (MM-LLM), vision based tasks such as generating images (Betker et al.) and video (OpenAI) can be performed, which can aid in the dissemination of LCA results (Decardi-Nelson et al., 2024). As a result, LLMs hold significant promise for developing a life cycle inventory (LCI) and summarizing and communicating LCA results, thereby improving the quality and decreasing the time necessary to complete an LCA (Cornago et al., 2023). However, the consequences of integrating LLMs with all stages and tasks of an LCA has not yet been explored.

Because it is not expected that LCA practitioners are acquainted with LLMs, an overview of LLM architecture and training is given in Section 2. A discussion of the potential uses of LLMs in LCA studies, from goal and scope definition to disseminating LCA results, is given in Section 3. By holistically considering the integration of LLMs into LCA studies, four perspectives are identified and described in Section 4, from using prompt engineering to improve LLM performance to integrating LLMs into LCA data management and processing frameworks. As shown in Fig. 1, there are numerous benefits for integrating LLMs throughout the LCA stages, but at the same time there are significant challenges and risks for integrating LLMs that need to be mitigated to realize the full benefits of these models, as described in Section 5. Finally, a sample output of three LLMs on a subset of LCA tasks is summarized in the

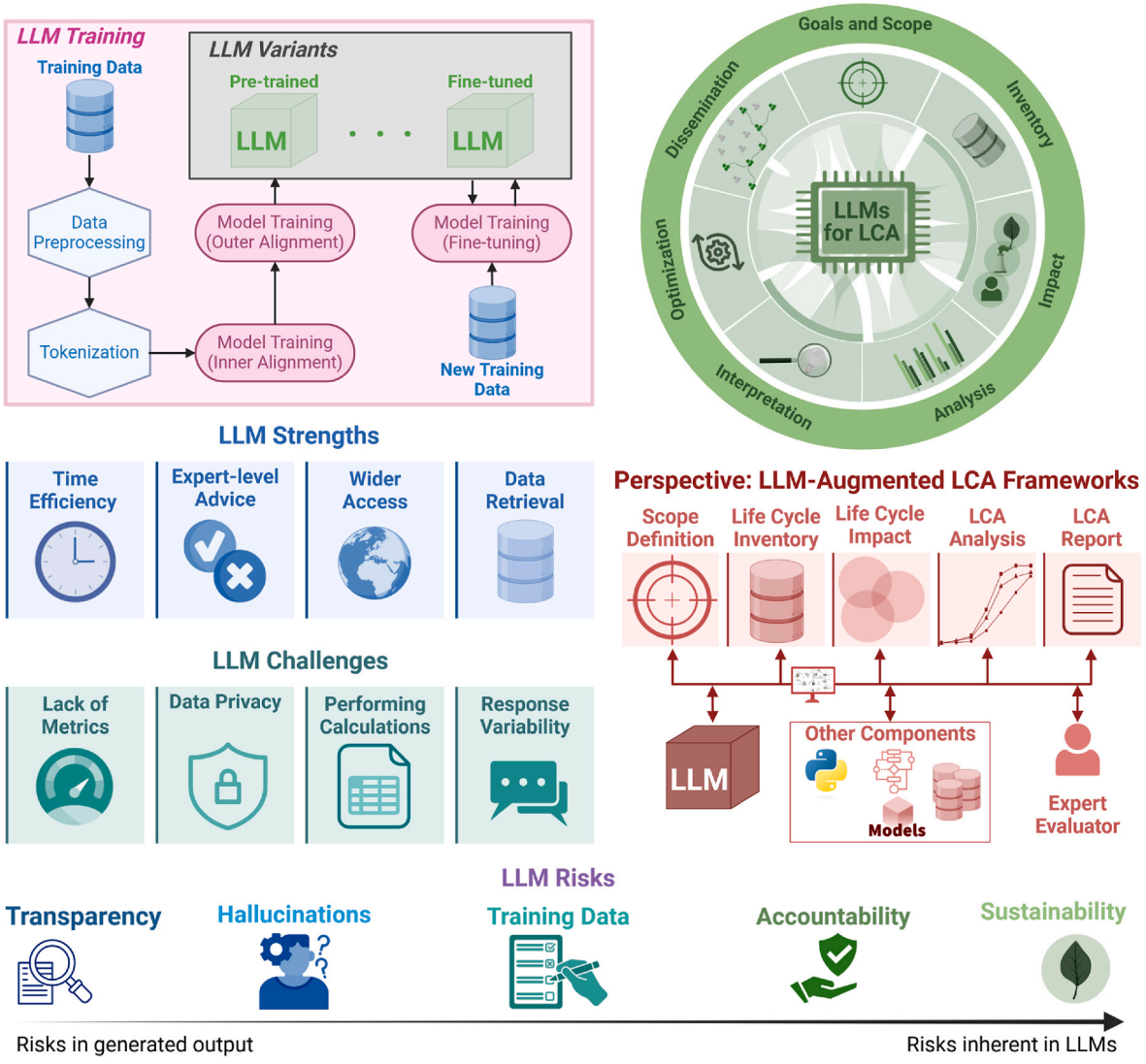


Fig. 1. Overview of the topics discussed in this study. An overview of LLMs, including training methods and model variants is given in Section 2. The strengths of LLMs and their application to a comprehensive selection of LCA tasks is given in Section 3. The challenges with using LLMs for LCA tasks are identified in Section 3, but methods to address these challenges are discussed in Section 4, culminating in the perspective of LLM-augmented LCA frameworks to increase LCA quality while spending significantly reduced time on the LCA. Section 5 discusses the risks of using LLMs for LCA tasks and potential mitigation methods.

Appendix, with a longer discussion of the quality of the output in the supplemental information.

2. Overview of large language models

This section is not meant to be a technical review of LLMs, but rather a brief overview that sketches the definition of LLMs, their architecture, training processes, and use. The various model architectures are sketched in Section 2.1, followed by a description of dataset collection and LLM training in Section 2.2, and the drivers of performance of LLM variants in Section 2.3. An overview of the training and use process is in Fig. 2. LCA practitioners interested in a more in-depth analysis of LLMs are encouraged to read these extensive reviews (Min et al., 2024), (Zhao et al., 2023).

2.1. Architecture

LLMs are typically based on a transformer architecture (Lin et al., 2022), which consists of two blocks, an encoder and decoder, as seen in the architecture section of Fig. 2. The encoder takes an input, or prompt, and maps it onto a target vector, which the decoder then translates into an output (Sutskever et al., 2014). LLMs are grouped into autoencoding models, autoregressive models, and sequence-to-sequence models, depending on which parts of the transformer architecture are used.

LLMs that only use a decoder are also called autoencoding models, which typically take in a user prompt and generate a response as an output. Common autoencoding models including GPT-4 from OpenAI (OpenAI, 2023), and Gemini from Google among others, and are used as the basis for popular chatbots like ChatGPT. When autoencoding models are trained, they attempt to predict the next token in a sequence given the previous tokens, in essence solving a very large conditional probability problem (Min et al., 2024). Decoder models are likely the most useful type of LLM to complete the tasks in Sections 3.1, 3.3, 3.4, and 3.5, as most of the tasks rely on generated output from LLMs.

Autoregressive models only use the encoder portion of the transformer, and are generally trained to guess the original series of tokens from a corrupted series of tokens (Min et al., 2024). While common encoder models such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019) can be used to generate text, it is more common to use them to understand and classify sentences (Sun et al., 2019). Other autoregressive models, like the open-source Llama-2 from Meta (Touvron et al., 2023), maintain chatbot functionality and will have use cases similar to the autoencoding models. Unlike the autoencoding models, there is limited opportunity for using autoregressive models for LCA tasks due to the fact that most of the tasks are generative in nature.

Encoder-decoder models are typically used in sequence-to-sequence (Sutskever et al., 2014) tasks, like machine translation, image and video captioning, question answering, and text summarization (Asadi et al., 2020). Some of the more common sequence-to-sequence models are BART (Lewis et al., 2019) and T5 (Raffel et al., 2023). While there are significant opportunities for this class of LLM, especially for text analysis and LCI data collection in Section 3.2, this type of model is not likely to be used in other sustainability tasks due to the lack of text generation.

While LLMs originally only produced text output, there is a growing trend towards developing MM-LLMs, which are able to produce many kinds of output, including images and video (Zhang et al., 2024). Notable MM-LLMs include Gemini, which improves the state of the art across 20 multi-modal benchmarks (Gemini Team et al., 2023), (Pichai), and Sora, which generates high-quality videos based on a text prompt (OpenAI). This type of LLM can be useful for LCA tasks that could involve image generations, such as those in Section 3.1, 3.4, and 3.5.

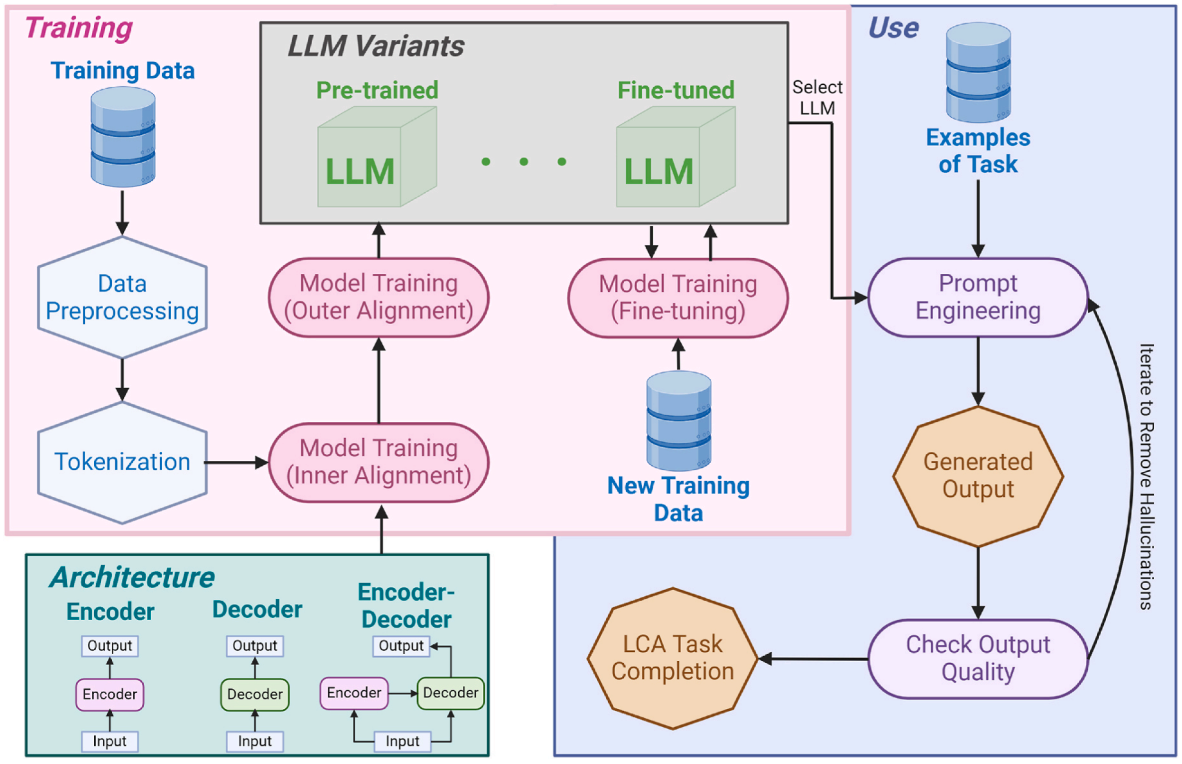


Fig. 2. Overview of development and use process for LLMs, from the different model architectures in Section 2.1, training the model in 2.2, and choosing a model variant for a task based on various criteria in Section 2.3. Model training begins with acquiring and preprocessing data, choosing an architecture, then training and aligning the model. The LLM can be fine-tuned to increase performance on a specific set of tasks based on a new training dataset, as discussed in Section 4.3. Once the LCA practitioner selects a model, prompt engineering and other techniques can be applied to improve the quality of LLM responses to LCA tasks as discussed in Section 4.1.

2.2. Training

As highlighted in Fig. 2, training LLMs begins with assembling the training dataset, which often involves acquiring enormous quantities of written text. Next, the dataset is processed by verifying the quality of the text and removing duplicates and private information to improve the performance of the model (Naveed et al., 2023). Once the dataset has been processed, it is often tokenized to decrease the computational cost of training.

The LLM training process consists of two phases, inner and outer alignment. In inner alignment, LLMs are trained to achieve the goals of the designer (Shen et al., 2023). This goal is typically achieved through minimizing the loss function by tuning the model parameters. This can be time-consuming and take significant computational resources, especially as the number of parameters in the model increases. Outer alignment of an LLM occurs when the training objective matches the goal of the human designer (Shen et al., 2023). Outer alignment is generally considered to be achieved if an AI model is considered to be helpful, honest, and harmless (Askell et al., 2021), and these properties are primarily developed through reinforcement learning with human feedback (RLHF) (Christiano et al., 2017), (Wolf et al., 2023). RLHF relies on the assumption that the values held by the humans in the RLHF loop are broadly shared with the rest of humanity, a feature which can result in some of the risks in Section 5. While most LLMs attempt to be outer aligned, some models, like Claude 2, are expressly developed to be as harmless as possible (Anthropic).

2.3. Variants

There are two main qualities, in addition to model architecture and training, for choosing an LLM for a task: the computational resource requirements and the fine-tuned model variants. LLMs are typically grouped into families of models which share a common architecture but are differentiated by the number of tunable parameters or the computational requirements. In general, LLMs with a larger number of parameters in the model outperform smaller models when trained on the same training data by having a response with greater complexity and verisimilitude (Srivastava et al., 2023), (Rosoł et al., 2023), (Brown et al., 2020). Therefore, it is recommended that the LCA practitioner choose the largest model if they have sufficient computational resources. For families of models with several fine-tuned variants on different training sets, a general rule of thumb is that LLMs that have been fine-tuned on a domain-specific task outperform a general LLM (Gururangan et al., 2020). Taking advantage of these findings. some organizations have pre-trained and fine-tuned LLMs on sustainability (Kaiwu), climate (Webersinke et al., 2022), or Intergovernmental Panel on Climate Change (Vaghefi et al., 2023) literature to improve model performance, but no one has yet built an LLM for LCA tasks.

3. The role of large language models in life cycle assessments

As discussed in the previous section, the appropriate choice of an LLM depends upon the tasks that need to be completed. While there are perspectives on the application of LLMs to environmental (Zhu et al., 2023a) and sustainability (Wang et al., 2023a) studies, there has not been an attempt to systematically assess the strengths and weaknesses of applying LLMs to every task and step in the LCA process. In this section, a set of tasks associated with LCAs are examined in the context of the advantages of LLMs, such as improving the accessibility of LCAs, increasing time efficiency, providing expert-level advice, and improving data management. In addition to the benefits of using LLMs, this section (1) demonstrates how LLMs are currently being applied, (2) discusses the challenges and risks associated with using LLMs in these tasks, (3) illuminates methods to overcome these shortcomings, and (4) provides examples for how LLMs will benefit LCA practitioners. Because LLMs are a novel tool for LCAs, to the best of our knowledge best practices for use have not yet been developed. Therefore, we encourage the LCA community to collaborate on standards for use.

In this section, a holistic approach is taken that considers LLM integration into goal and scope definition in Section 3.1, LCI in Section 3.2, life cycle impact assessment (LCIA) in Section 3.3, life cycle impact analysis, interpretation, and optimization in Section 3.4, and LCA results dissemination in Section 3.5. Through this analysis, the areas of opportunity, challenge, and risk throughout the entire LCA process are identified, and summarized in Fig. 3. The benefits of using LLMs for LCA tasks are summarized in Table 1. A baseline level of LLM performance on some LCA tasks can be found in sections S3-S5 of the supplemental information.

3.1. LLMs for goal and scope definition

Currently, some promise is shown for applying LLMs to tasks in the goal and scope definition stage, in part due to the expertise necessary for the LCA practitioner to make high-quality choices, including selecting the goal of the project, the LCA type used, the system boundary, and the functional unit (Cornago et al., 2023). The ramifications of LLM use for each of those four major choices are explored throughout the remainder of the subsection. Example conversations with three state-of-the-art LLMs and MM-LLMs on some goal and scope tasks are included in the supplemental information, though the responses are summarized in the Appendix.

The challenge in defining a goal for an LCA lies in the inherent subjectivity of the goal-setting process. For example, the LLMs in the supplemental information define the goal as either providing the decision-makers with objective environmental and socio-economic information, minimizing greenhouse gas emissions, or comparing three potential land use pathways. An LCA practitioner deciding on the goal will inevitably justify their decision in some capacity, perhaps by considering relevant factors from primary data collection, precedent, literature review, or their expertise. A summary of LLM justifications for these goals is in the Appendix.

Because the choice of goal affects the type of LCA method used (Rebitzer et al., 2004), and as some goals can conceivably result in the appropriate use of multiple LCA types, choosing an appropriate LCA type can be a difficult decision. An LLM trained specifically on LCA data would be able to have expert-level recommendation capabilities, which can compare and contrast context and problems without the need for an extensive literature review by the LCA practitioner. Until such an LLM is developed, it may be more practical to leverage latent rules of thumb for choosing an LCA type using few-shot learning, as mentioned in Section 4.1. While this may not save time for experts, it may be broadly useful to novices who do not have considerable experience in LCA studies. This enables an LLM to make LCAs more accessible to a wide range of practitioners, bridging the gap between sustainability concepts and non-experts.

There is both immense potential and elevated risk in using LLMs and MM-LLMs to generate a system boundary. Building a system boundary can take place in two steps, first gathering relevant process information and then drawing the system boundary. First, selecting appropriate boundaries can require a great deal of LCI data (Reap et al., 2008a). For this task, LLMs have promise given their ability to efficiently summarize data, as discussed in Section 3.2. LLMs can also be used to draw LCA system boundaries in the near future, as they have already shown promise in generating chemical process flowsheets (Hirtreiter et al., 2024), a similar task that maps input and output flows in a system. These system boundary representations could be improved if a text-based representation for system processes can be developed from scratch or adapted from chemical process design (White, 2023) because LLMs excel at producing text-based output. If the LCA practitioner would prefer an image-based LCA system boundary, MM-LLMs can use the gathered system boundary data to generate an image. If an LLM makes a mistake in the system boundary, the LCA practitioner can give feedback to the



Fig. 3. The strengths, challenges, opportunities, and risks of using LLMs for LCA tasks. The strengths and challenges are primarily discussed in Section 3, while the opportunities are discussed in Section 4 and the risks in Section 5.

LLMs through few-shot learning to correct these errors and iteratively improve upon the system boundary. One potential risk is that reliance on LLMs and MM-LLMs for this step may limit the understanding of the LCA practitioner about the system of interest, which may lead to improper or lower-quality analysis of the LCA results. If the LCA practitioner engages with the LLM through chain of thought reasoning as discussed in Section 4.1, the practitioner could gain a comprehensive understanding of the system while also verifying the justification used by the LLM to draw the system boundary.

The choice of an appropriate functional unit in LCA is difficult (Cooper, 2003), as it requires the consideration of the lifetime, performance, and dependencies of a system, which generally requires experience and/or expertise (Cornago et al., 2023). While some LCA domains may have a standardized functional unit developed through best practices, others may not (Marinković, 2017). For the tasks in the Appendix, some LLMs were able to generate a list of potential functional units in a zero-shot context. However, the functional units gravitate towards generic functional units that do not necessarily incorporate the lifetime

Table 1
Benefits derived from using LLMs for specific LCA tasks.

Section	Task Name	Wider Access	Expert-Level Advice	Data Retrieval	Time Efficiency
Section 3.1	Goal Definition		X		X
	LCA Type Selection	X	X		
	System Boundary		X	X	X
	Functional Unit		X		
Section 3.2	LCI Data Collection			X	X
	LCI Data Estimation			X	
	Data Management			X	X
Section 3.3	Selection of LCIA Method	X	X		
	Highlighting Impact Categories	X	X		
	Life Cycle Impact Calculation				
Section 3.4	LCA Result Analysis		X		X
	LCA Result Interpretation		X	X	X
	LCA Result Optimization	X	X	X	X
Section 3.5	Communicate LCA Results		X		
	Sustainable Workforce Development	X			
	Disseminate LCA Findings	X	X		X

and performance of the system, which could be addressed using few-shot learning as discussed in Section 4.1. To enable the adoption of technological tools for LCAs and facilitate comparisons between LCAs, we encourage environmental scientists to develop a consensus on functional unit definitions. In the near future, LLMs may generate complex functional units that balance competing goals better than an LCA expert.

LLMs may be able to generate output that replicates or exceeds the quality of an LCA practitioner's work for the goal and scope definition. There are two challenges for integrating LLMs into this step of the LCA workflow: 1) there is a lack of methods to compare the quality of LCA practitioner and LLM responses to goal and scope tasks, and 2) the presence of hallucinations in the generated output. To better gauge the progress made by LLMs on LCA tasks, it is imperative that benchmarks are developed or adopted to quantify both novice LCA practitioner, expert LCA practitioner, and LLM performance on various goal and scope definition tasks, as discussed in Section 4.3. The second challenge is primarily addressed using the various prompt engineering techniques discussed in Section 4.1, or by selecting a higher quality or more specialized LLM. The primary risk for integrating LLMs into this step of the LCA workflow is that the LCA practitioner does not accept responsibility for the content produced by the LLM. This risk can be addressed by ensuring that the LCA practitioner knows the ethics for using LLM-generated content in LCAs, as discussed in Section 5.4. By integrating LLMs into goal and scope definition tasks, the LCA practitioner can decrease the time required to complete these tasks without necessarily making a sacrifice in LCA quality. Moreover, LLMs make LCAs more accessible to non-experts and enable the practitioner to gain a more comprehensive understanding of the system of interest.

3.2. LLMs for life cycle inventory

LLMs have great promise for compiling and evaluating the LCI for an LCA because of their success in natural language processing tasks (Min et al., 2024). Currently, LLMs can only describe the LCI of common products (Cornago et al., 2023), primarily because there is sufficient literature on those products included in the training data. This capability is expected to be significantly expanded in the near future, as LLMs are democratizing LCI data collection, are being integrated with machine learning (ML) models to estimate LCI data for novel materials, and are being used as query processors to easily retrieve LCI data from the literature or a database.

Several projects are currently leveraging LLMs to gather LCI data. In the TianGong project (TianGong), LLMs are being used to process databases, material safety data sheets, and published LCA results to develop a comprehensive LCI database. In this project, an LLM processes the text and identifies locations in the text likely to contain LCI data, in terms of both processes and impact factors. The data is then verified by humans with reference to the specific location in the underlying report and then standardized before being entered into an LCI database. Furthermore, AutoPCF (Luo et al., 2023a) uses LLMs to extract emissions impact factors for the raw material extraction and manufacturing process phases from any LCI data source. Then, the LCI data is matched to emissions factors using a semantic-based emission factor matching deep learning model. It isn't strictly necessary that a language model be large to achieve near-expert performance on identifying impact factors, with the Flamingo project using deep learning models to return impact factors from a database using text matching (Balaji et al., 2023) and using BERT-derived models to estimate scope 3 emissions (Jain et al., 2023a). Similar projects exist for supplementing LCI database access through an AI agent, such as in the WISER project (WISER), (Bekavac and Volk, 2024). By enhancing LCI data management with LLMs, the quality of LCAs can be improved while LCA practitioners spend less time on LCI data collection.

In the event that the LCI data is not found in the literature or the database, it becomes necessary to estimate these parameters. At first glance, it appears feasible to directly prompt LLMs to generate LCI data, especially for novel materials for which the data may not exist (Turhan, 2023). Using LLMs to generate LCI data is fraught with risks, as it is exceptionally difficult to explain how the LLM generated the data and hard for the LCA practitioner to verify the accuracy of the data, as explained in Section 5.1. Therefore, it is instead recommended to use LLMs to develop a list of features for novel materials and use those features as an input to existing ML models that estimate LCI data. Using ML models is desirable because they offer high performance, can be assessed on uncertainty prediction methods (Amini et al., 2020), and are more explainable (Ghoroghi et al., 2022). This method has established state-of-the-art performance for predicting chemical and physical properties of crystals from a text description (Rubungo et al., 2023) and could easily be extended into the LCI domain. By integrating ML models into the LLM data management pipeline, the advantages of accuracy and explainability from ML models can be combined with the time savings of using LLMs to achieve high-quality LCI data estimation in a significantly reduced timeframe, as mentioned in Section 4.2.

More broadly, integrating LLMs with data management techniques using vector databases (VDB) can enable LLMs to search through a precompiled set of documents and reference the original documents in generated responses without relying on any one single LCI database (Zhou et al., 2024). The use of VDBs can also overcome potential shortcomings of the LLM training process. Because LLMs can "remember" and repeat sensitive information from the training dataset in other contexts, sensitive data should be kept in VDBs, as the LLM cannot divulge information from VDBs in external contexts. This is primarily relevant when LCAs are conducted on products from mutually competitive firms, wherein it is advantageous to the firm to hide information on specific flows, processes, and supply chains from the competition (Kuczenski et al., 2017). Using LLMs enhanced with databases, if adapted to LCA tasks, could decrease the time required to gather LCI data and impact factors from days down to minutes. In the traditional method, the LCA practitioner may use an LCI database like EcoInvent (Wernet et al., 2016) or manually search through the literature, which can be time-consuming tasks. Instead, the practitioner could

prompt an LLM in their native language to query a database (Maddigan and Susnjak, 2023), ensuring that a user does not need to learn a special programming language to access the data contained therein (Petroni et al., 2019). The LLM could then return several processes, each corresponding to a different manufacturing method or locale while citing the specific location in the file from which the data was acquired. In the event that there are multiple feasible impact factors for a given process, an LLM may be able to verify the accuracy of the impact factor and provide guidance on the contexts in which each of the impact factors should be used, enabling the practitioner to improve the quality of the LCI data while simultaneously providing good uncertainty bounds for a sensitivity analysis. Further perspectives on LCI data management can be found in Section 4.2, and more information on LLM-integrated LCA frameworks can be found in Section 4.4.

Integrating LLMs into LCI collection pipelines has several advantages over traditional LCI data collection methods. These include faster extraction of relevant LCI data from the literature and databases and using text descriptions and integrated ML models to predict impact factors. The use of LLMs to improve the quality, accuracy, and transparency of LCI data has the potential to set the standard for LCI databases in the near future. Moreover, the use of human verification of data extracted by the LLM should ensure that no hallucinations or inaccuracies are entered into the LCI database or migrate their way into an LCA. Continued collaboration between LCA practitioners and LLMs will likely become the norm for LCI data collection and processing due to the inherent advantages in efficiency and transparency.

3.3. LLMs for life cycle impact assessment

In this stage of the LCA there are three main tasks that could benefit from the integration LLMs: choosing an LCIA method, highlighting certain impact categories, and completing the life cycle impact calculation. Of the three stages, this stage is least suited for using LLMs, but could benefit from integrating LLMs into an LCA pipeline as discussed in Section 4.4.

To conduct the LCA, first the LCIA method must be chosen. These methodologies have their strengths and weaknesses, as they make different assumptions about the location of the environmental impacts, as well as the impact factors of different material flows (Borghesi et al., 2022). To test the promise of using LLMs to select and justify an LCIA methodology, three LLMs were prompted in a zero-shot learning context, and the responses are recorded in the supplemental information. While the output sounded authoritative in some contexts, some LLMs included hallucinations, suggesting that prompt engineering as discussed in Section 4.1 or an LCA-specific LLM as mentioned in Section 4.3 could yield improved performance.

LLMs show promise for a related LCIA task. As some LCIA methodologies contain tens of impact categories, it may be unclear to the user what harms are being measured by what category. LLMs can use their knowledge base to summarize the different impact categories and explain to the user the science behind each impact category and the use cases for highlighting that category in an LCA. This use of LLMs may enable the LCA practitioner to focus on a few key LCIA indicators in an LCA, potentially aiding in the sustainable optimization of the system and increasing the clarity of the reported results.

Because LCI data is commonly stored as a matrix and because LLMs are trained to handle a sequence of inputs, it does not make sense for a LLM to calculate the LCIA. While it is possible for an LLM to read in the matrix line by line and do calculations, this is not an ideal approach. Instead, an LCA practitioner would likely benefit from an automated LCA pipeline that integrates LLMs and other programs, relying on a dedicated LCIA program like OpenLCA (GreenDelta) or SimaPro (SimaPro) to complete these calculations, as discussed in Section 4.3.

3.4. LLMs for analyzing, interpreting, and optimizing LCA results

LLMs hold great promise for aiding in the analysis, interpretation, and optimization of LCA results. LLMs can save the LCA practitioner time, increase the accessibility of life cycle optimization (LCO), and provide policy implications. While there are challenges surrounding hallucinations and responsibility described in Section 5, these can be alleviated through the use of LLMs specifically trained on LCA data, by integrating LLMs into LCA data management frameworks, and through various prompt engineering techniques.

The primary issue with analyzing the LCA results is the preponderance of data resulting from multiple pathways and the processes within each pathway. Analyzing this data can be time-consuming even when using dedicated software, but LLMs offer a promising method of quickly and accurately analyzing LCA results. The benefits of LLM-enabled results analysis include the identification of environmental hotspots (Cornago et al., 2023) and other irregularities in the LCIA data. This integration minimizes the risk of hallucinations, as the results can be compared instantly to similar literature to identify any irregularities and offer numerous opportunities to the LCA practitioner to rectify hallucinations. For example, once the hotspots and irregularities are identified, a component breakdown can be conducted to identify the primary causes. The LLM could then recommend further analysis, including a list of data-driven parameter values for prospective processes to use in a sensitivity analysis. The decision to use these variables lies with the LCA practitioner, as they should take responsibility for the content produced by LLMs, as discussed in Section 5.4. From here, the sensitivity analysis could be passed off to a dedicated LCIA program as recommended in Section 3.3 as part of an LLM-integrated LCA pipeline described in Section 4.4. By the end of this process, the hallucinations would likely be identified and corrected by the LCA practitioner, while quickly validating and verifying the important findings. These above tasks can be further augmented by using MM-LLMs to improve the visualization of the results.

Once the results are analyzed they can be interpreted to identify the key results and policy implications. The baseline ability of LLMs to identify the key results depends on their performance on data analysis tasks, and their ability to understand what makes a result key in the context of an LCA. As LLMs are quickly improving in their ability to reason mathematically (Gemini Team et al., 2023), this capability will continue to improve in the near future and offer high-quality insights and findings based on LCIA data. More caution should be applied when using LLMs as the basis of policy recommendations, though there are several strategies to ensure that the policy implications are less harmful. As output from an LLM can appear authoritative even if it is incorrect, it is strongly recommended that the LCA practitioner verify the policy implications produced by the LLM to ensure that they are harmless and less biased (Rillig et al., 2023). To aid in the development of a less harmful LLM in the context of an LCA, an LCA-specific LLM should be developed, such that it can be fine-tuned to give a higher weight to “harmlessness”, as referenced in Section 4.3. Moreover, if the LLM does not know or has a high uncertainty about the potential policy implication, then it should report that information to the LCA practitioner so that they can appropriately verify the implications contained in the output and reduce potential harms from the policy. While efforts can be taken to reduce bias in the content generated by LLMs, as detailed in Section 5.2, this is an issue that cannot be avoided. Finally, it also may be useful to construct some benchmarks to evaluate the quality of policy implications (Nicholson, 1997) to gain a better understanding of the potential and shortfalls of using LLMs for policy writing, as discussed in Section 4.3.

Current methods of LCO can be improved through LLM integration. A reliance on computational optimization using either deterministic or heuristic solvers restricts the ability to conduct LCOs to those LCA practitioners with the requisite mathematical and programming knowledge. By leveraging the ability of LLMs to suggest potential

process improvements (Zhuang and Wu, 2023) and revise product design to achieve specific environmental objectives (Baduge et al., 2022), LCA practitioners can quickly identify changes to supply chains to ameliorate the environmental hotspots. This capability can be strengthened by using an LLM trained to recognize patterns and associations that are indicative of sustainable design practices (Turhan, 2023). Finally, integrating the LLM into an LCA data pipeline enables fast iteration through potentially optimal design by quickly gathering appropriate LCI data and impact factors as discussed in Section 3.2. While this approach to LCO may lack the rigor associated with using optimization theory, it may enable LCA practitioners without a strong mathematical background to identify more sustainable process designs and suggest policies to enable the adoption of those improvements. Eventually, this makes sustainable design ideals more accessible and could accelerate the adoption of green supply chains.

3.5. LLMs for disseminating LCA results

Once an LCA has been completed, the next step is to disseminate the results to the relevant stakeholders and to the public at large. While traditional LLMs can only do this through text, MM-LLMs can also share these results as images or videos, broadening the potential audience for the LCA results. As there has already been research into the abilities of LLMs to engage in writing (Salvagno et al., 2023), the focus in this study is instead turned towards the use of LLMs to disseminate sustainability findings. LLMs have great promise to radically transform disseminating results from LCAs, and given the critical nature of climate crisis (IPCC, 2023), it offers a potentially transformative expansion in educating the general public on the climate crisis broadly and sustainability crises locally. This is a multi-scale opportunity, from using an LLM to explain LCA results to a single user, to expanding the accessibility of LCAs, thereby training and educating a more sustainable workforce, to summarizing LCA findings in reports, including reports on environmental, social, and governance (ESG) targets. At the individual level, LLMs are likely to function more as search engines, giving summaries of sustainability practices to users, while at the larger scale they hold the potential to help write documents read by thousands or millions.

As LLMs have the capability to replace search engines (del et al., 2023), it is probable in the near future that LLMs will be used as a tool to communicate LCA results with individual users (Wang et al., 2023a). The user may give a prompt without much context, such as "What can I do to reduce energy usage at home?", but may also be context-specific, for example, "Can I recycle this plastic tray?" alongside an image of the object. While the hope is that an LLM will provide a factual, nuanced, and honest response based on relevant LCA results, there is always the risk of an LLM hallucination, which could be minimized by training LLMs on climate or sustainability literature and ensuring that appropriate environmental viewpoints are adopted during outer alignment.

By increasing the accessibility of LCAs, LLMs hold significant promise for workforce training and development. Compared to traditional methods of education, LLMs hold several advantages, particularly in engaging students and developing personalized learning plans (Kasneci et al., 2023). This has led some to describe LLMs as a transformative shift in education, opening up sustainability education to a far broader audience (Wang et al., 2023a), (Haluza and Jungwirth, 2023), with great promise being shown in medicine (Safranek et al., 2023) and higher education (Meyer et al., 2023) among other disciplines. While a major concern of using LLMs for workforce development includes overreliance on the models on part of both students and teachers resulting in a lack of understanding and expertise (Kasneci et al., 2023), this could be mediated through chain-of-thought reasoning as discussed in Section 4.1. These features enable an LLM to make LCAs more accessible to a wide range of practitioners, bridging the gap between sustainability concepts and non-experts.

LLMs may also be used to help draft reports intended for the general public that draw on findings from LCAs. The most prominent example of this would be using an LLM to help draft a report highlighting progress towards ESG goals. ESG goals are a set of metrics that act as a proxy to measure the sustainability of a corporation's actions (Widyawati, 2020), as part of the corporate social responsibility framework (Lindgreen and Swaen, 2010). LLMs may improve the accuracy, efficiency, and accessibility of corporate sustainability reporting, improve the quality of analysis of non-financial data, and make information more accessible to stakeholders (de Villiers et al., 2023). However, like with the other facets of sustainability communication, there may not be an improvement in transparency and accountability because LLMs are prone to bias and error (de Villiers et al., 2023), though this risk can be minimized by using an LCA-specific LLM with expert-level recommendation capabilities as mentioned in Section 4.3.

4. Strategies for enhancing the integration of large language models into life cycle assessments

In addressing the integration of LLMs into LCAs, several challenges emerge, notably hallucinations where LLMs produce plausible yet incorrect information (Huang et al., 2023) because of the models' lack of domain-specific knowledge. Trained predominantly on public datasets, LLMs may falter in answering specialized questions beyond their training scope (Penedo et al., 2023). Furthermore, their knowledge may quickly become outdated, given the dynamic nature of real-world information. The high cost of mainstream LLM fine-tuning and integration also presents a barrier, rendering it impractical for non-tech entities to develop or customize such models due to resource constraints (Musser, 2023). Additionally, LLMs exhibit a tendency towards catastrophic forgetting, posing significant challenges for their use in continuously evolving fields like LCA (Luo et al., 2023b). This section proposes strategies to mitigate these limitations through prompt engineering in Section 4.1, VDB integration in Section 4.2, fine-tuning for the development of expert-level recommendations in Section 4.3, and the development of LLM-augmented LCA frameworks in Section 4.4. It also reviews pioneering applications of LLMs in sustainability that resonate with these approaches and can be leveraged to improve the quality of LCAs. Fig. 4 visually elucidates the strategies outlined and their components towards advancing LLM applications in LCA.

4.1. Prompt engineering for improved LLM responses

As evidenced in the Appendix, the quality of responses of LLMs on LCA tasks could be improved. To realize potential performance increases and reduce the incidence of model "hallucinations", the LCA practitioner can use a collection of techniques that are commonly referred to as prompt engineering. This involves crafting specific instructions, or prompts, to guide the output of LLMs, enabling task-specific adaptability without modifying underlying model parameters. Unlike traditional methods requiring retraining or extensive fine-tuning mentioned in Section 4.3, prompt engineering offers a nuanced approach to adjust model outputs across various tasks and domains effectively (Sahoo et al., 2024), (Chen et al., 2023a). There are three main types of prompt engineering: few-shot learning, chain of thought (CoT) reasoning, and information retrieval processes. These techniques are not mutually exclusive.

To understand the effectiveness of few-shot learning, it is important to first differentiate it from zero-shot learning. The prompts included in the supplemental information are zero-shot learning prompts because the LLM is asked to complete a task without any examples (Xian et al., 2019). Zero-shot learning prompts can be low-quality, like the prompts in the supplemental information, but when done properly zero-shot learning prompts can elicit high-quality responses to novel tasks. Few-shot learning, on the other hand, introduces multiple examples within prompts to enhance the model's grasp of task nuances in LCA-specific contexts (Brown et al., 2020). This approach requires significant space in the prompt for examples, potentially limiting its use

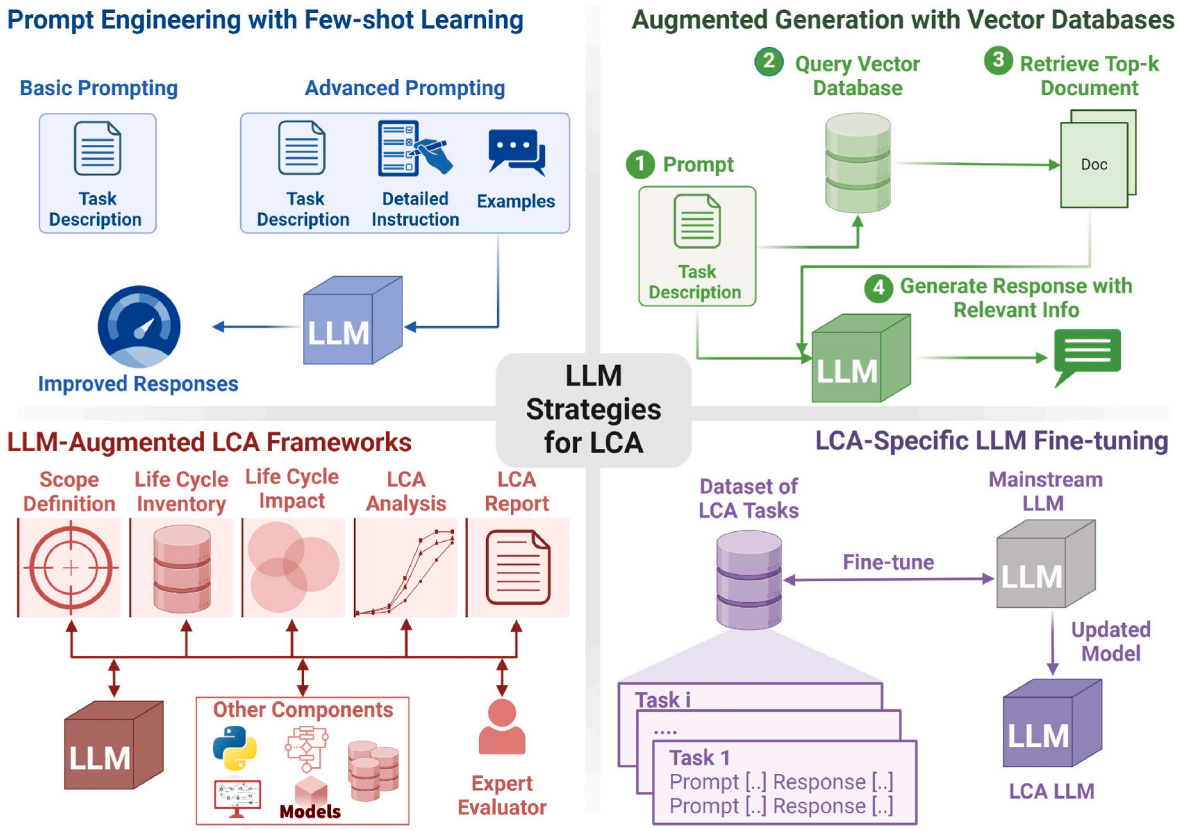


Fig. 4. Overview of perspective strategies for enhancing LLM integration into LCA, delineating key approaches to overcome challenges in applying LLMs to LCAs, including prompt engineering, vector database integration, fine-tuning, and development of LLM-augmented frameworks.

with extensive texts that are longer than the context window of the LLM. This limitation can be addressed through expanding the size of the content window, which can require additional fine-tuning (Chen et al., 2023b). The choice and arrangement of these examples are critical, as model output can be skewed by biases, including the propensity for common terms and tasks (Brown et al., 2020). Few-shot learning typically enables an LLM to generally achieve higher performance on a task when compared to the performance of a zero-shot model (Min et al., 2022). Therefore, it is recommended to use few-shot learning for the many LCA tasks involving text generation throughout Section 3 because it is a much more efficient process than fine-tuning and offers substantial performance improvements over using zero-shot learning.

Furthermore, LLMs often struggle with complex reasoning tasks (Wei et al., 2023), which form the basis of LCAs. To address this challenge, the CoT prompting technique has been developed. CoT prompts can range from the simple "think step-by-step" to complex curated prompts (Zhang et al., 2022) through methods such as automatic CoT generation (Zhang et al., 2022), consistent CoT sampling (Wang et al., 2023b), and logical trees (Yao et al., 2023a), (Yao et al., 2023b). Using this prompting technique can enable an LCA practitioner to better understand the justifications given by the LLM for LCA design decisions, especially for any task in the expert-level advice column in Table 1.

Additionally, prompt engineering can reduce model-generated errors, or "hallucinations," through strategies incorporating information retrieval from curated databases (Lewis et al., 2021) and automatic verification processes (Dhuliawala et al., 2023), which will be addressed in Section 4.2. These qualities are especially desirable for LCA studies, as there are significant time-saving benefits to be realized, primarily for LCI data collection as discussed in Section 3.2.

As seen, these prompt engineering approaches represent the most accessible short-term solutions to bolster LLM performance across LCA tasks. We advocate for the LCA community to actively participate in defining diverse, representative tasks and examples, as discussed in greater detail in Section 4.3. Sharing these tasks and examples will not only benefit prompt engineering efforts but also facilitate the fine-tuning of future LCA-specific LLMs.

4.2. *LLM-enhanced data management for efficient data retrieval*

There are two main advantages for integrating LLMs into an LCA data management ecosystem. First, this approach can help address the presence of hallucinations in LLM-generated output. Second, this approach is currently being used to develop LCI databases and can be easily extended to incorporate other sources of LCA data, such as a corpus of published LCA texts.

One major challenge with LLM-generated output in general, and the responses in the Appendix in particular, is the presence of hallucinations, which are described in detail in Section 5.2. While there are many causes for hallucinations in general, in the context of LCAs it is likely that the hallucinations stem from limited domain knowledge and outdated knowledge. To address this challenge, integrating updated databases into LLMs emerges as a viable and cost-efficient solution. VDBs enhance LLMs by providing in-context retrieval of domain-specific documents, thereby bolstering data privacy through local storage and local or secure API LLM generation (Han et al., 2023). This integration enables the efficient retrieval of documents highly relevant to user queries, drawing from prebuilt databases containing user-provided and online data sources, such as LCA journal articles. Subsequently, these documents are provided as input to augment LLM responses, infusing outputs with current and specialized knowledge. This strategy ensures LLMs produce enriched outputs in specialized areas, bypassing the need for extensive LLM training while ensuring the utilized sources are referenced for verification (Zhang et al., 2023a), (Jing et al., 2024). VDBs are useful throughout the LCA process, from being able to cite

peer-reviewed precedents for difficult goal and scope definition tasks in Section 3.1 and to verifying data sources for LCI tasks in Section 3.2.

Innovations in LCA VDBs are exemplified by the TianGong AI project, which integrates GPT LLMs with sustainability databases through a retriever-answer generator ecosystem. This system facilitates efficient data retrieval by linking a ChatGPT plugin and customizable chat interfaces to external knowledge bases (TianGong). Furthermore, additional efforts have been made to supplement the EcoInvent LCI database with retrieval augmented generation (RAG) to aid the LCA practitioner in identifying appropriate processes and flows (Bekavac and Volk, 2024). RAG and VDBs can enable transparency throughout the production chain, offering increased insight into the emissions for each step of the production process. Parallel efforts in other fields have led to the creation of custom VDB frameworks that enable the deployment of both online and offline LLMs (Zhou et al., 2024), (Bang et al., 2023). Nonetheless, LCA practitioners face multifaceted challenges that warrant careful consideration for integrating LLMs with custom VDBs. The limitations imposed by journal publishers on text mining significantly restrict the inclusion of comprehensive paper content, often confining the scope of the included data to smaller text segments. Compounding these challenges is the LLM-specific nature of these databases, rendering VDBs created by models like Gemini incompatible with those from GPT (Han et al., 2023). Moreover, the burgeoning volume and indeterminate quality of LCA literature calls for the development and implementation of advanced algorithms for more effective filtering, indexing, and retrieval, to ensure the efficient management of this extensive data pool. Some of these challenges could be addressed through the development of an open-source LCA-specific LLM, as discussed in Section 4.3.

4.3. LCA-specific LLMs for expert-level recommendations

While VDB integration enables LLMs to retrieve narrowly focused LCA information in response to user prompts, aligning LLMs with the views of LCA experts may necessitate fine-tuning and the development of specialized datasets and benchmarks. Fine-tuning, a process where LLMs undergo additional training to specialize in specific domains, equips these models with a broader contextual understanding and improves the performance of these models on domain tasks (Li et al., 2019). Yet, refining mainstream LLMs demands significant computational resources, as the adjusted model retains the size of the original LLM (Jeong, 2024). For these reasons, it is unlikely that an LCA-specific LLM will be developed in the near future, requiring LCA practitioners to rely on the proxy benchmarks as listed in Table 2 to select an appropriate LLM for the task.

Another hurdle, besides training costs, in tailoring LLMs for LCA applications involves the need for extensive and diverse datasets for fine-tuning, necessitating collaborative cross-institutional and community-wide data collection efforts. To establish comprehensive datasets to evaluate LLM performance and enable the development of LCA expert-level LLM recommendations, an LCA-specific set of benchmarks and datasets will need to be developed. Generally, the datasets for LLM benchmarks derive from publicly available questions (Hendrycks et al., 2021a), though others are painstakingly created by a team of human workers (Cobbe et al., 2021). For LCAs, the former is not an option. Thus, the dataset curation process should start by establishing quantitative measures of the quality of LCA tasks. While the process of LCA is defined in ISO standards (Finkbeiner et al., 2006), and while there are practical guides with best practices for conducting LCAs (Guinée et al., 2001), (Tokede and Traverso, 2020), to the best of our knowledge there are no agreed upon quantitative benchmarks that can differentiate a high-quality LCA from a low-quality LCA. Therefore, developing datasets and an associated set of benchmarks necessitates collaborative cross-institutional and community-wide discussion and consensus-building. Once the benchmarks are agreed upon, the next major task is to label the data, which can be time-consuming. In the near term, it is recommended that LCA practitioners who develop examples for few-shot learning make these examples publicly available so that other researchers can use them to improve LLM performance. In the medium term, it is recommended that teams of LCA experts build large, varied, high-quality datasets for LCA tasks. It is critical to note that LLM research underscores a power-law relationship between dataset size and model performance, hinting at the critical role of expansive datasets in establishing the efficacy of domain-specific LLMs (Kaplan et al., 2020). The ultimate objective is to develop an LCA-LLM that not only minimizes hallucinatory outputs, but also precisely discerns the extent of its knowledge and the implications of data extracted with VDBs. This LLM could have the capability to provide expert-level recommendations for several of the more complicated LCA decisions, especially for the tasks listed in Sections 3.1 and 3.4.

Because this process will be time-intensive, in the near term LCA

Table 2
List of LCA-specific tasks in need of benchmarks with potential information for labels and features, and a list of similar LLM benchmarks as a proxy.

LCA Task Name	Features	Labels	Similar Benchmarks
Goal definition	LCA context	LCA goal	WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019)
LCA type selection	LCA goal	LCA type	WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019)
System boundary	LCA context, goal, type	LCA system boundary	WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019), MMMU (Yue et al., 2023)
Functional unit	LCA context, goal, type, system boundary	LCA functional unit	WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019)
LCI data collection			Classification Metrics (Powers, 2020)
LCI data estimation			Classification Metrics (Powers, 2020)
Data management			Classification Metrics (Powers, 2020)
Selection of LCIA method	LCA goal, type, functional unit, system boundary	LCIA method	WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019)
Highlighting impact categories	LCA goal, type, functional unit system boundary, LCIA method	LCIA impact categories	WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019)
Life cycle impact calculation			GSM8k (Cobbe et al., 2021), MATH (Hendrycks et al., 2021b)
LCA result analysis			WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019), GSM8k (Cobbe et al., 2021)
LCA result interpretation			WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019), GSM8k (Cobbe et al., 2021)
LCA result optimization			WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019), GSM8k (Cobbe et al., 2021)
Communicate LCA results			WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019)
Sustainable workforce development			WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019)
Publish LCA findings			WinoGrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019), MMMU (Yue et al., 2023)

practitioners could use existing LLM benchmarks as a proxy for evaluating LLM performance on LCA tasks, particularly those focused on data management, mathematics, and reasoning. Many fields have developed benchmarks to evaluate LLM performance, such as linguistics, biology, physics, and software development (Srivastava et al., 2023). These datasets are useful for evaluating LLM performance on domain-specific problems like math in GSM8k (Cobbe et al., 2021) and MATH (Hendrycks et al., 2021b), more abstract commonsense reasoning problems like WinoGrande (Sakaguchi et al., 2019) and HellaSwag (Zellers et al., 2019), or a range of diverse, multi-modal subjects in MMMU (Yue et al., 2023). Table 2 gives a concise overview of the different tasks that could benefit from an LCA-specific dataset and gives a list of potential data sources upon which benchmarks could be extracted.

4.4. LLM-integrated LCA frameworks for automated LCA

The ultimate goal of integrating LLMs into LCA is the development of an LCA ecosystem that seamlessly integrates LLMs into every step of the LCA process. This scenario envisions using LLMs as a plug-in to existing LCA software, enabling a variety of external LLMs with different specialties to be used in various points throughout the LCA process. While the development of such a software suite is likely years away, it will best realize the benefits of using LLMs to conduct the LCA tasks listed in Section 3. Such integration will massively reduce the time necessary to complete an LCA and likely establish a more uniform quality of LCA. To enable an LLM-integrated LCA pipeline for automated LCA, several challenges need to be addressed. First, the sequential and mathematical reasoning of LLMs needs to be improved, though promise has been shown for modeling frameworks that comprise a subset of LCA tasks. Second, the complexities inherent in LCAs need to be tamed, which will rely on the developed expertise of LCA practitioners.

Despite their advanced capabilities, LLMs exhibit limitations in performing tasks requiring mathematical and sequential reasoning, as is often required in LCAs (Sahoo et al., 2024), (Chen et al., 2023a). The development of frameworks that deconstruct the complex steps of LCAs can significantly enhance the efficiency and effectiveness of LLMs in this field. Such frameworks enable LLMs to methodically review literature, extract pertinent information, inform LCA decisions, execute scripts for calculations, and ultimately summarize and disseminate findings. A notable initiative in this arena is AutoPCF, which, though focused on the simpler “cradle-to-gate” life cycles for product carbon footprint (PCF) analysis (Luo et al., 2023a), demonstrates the potential of integrating mainstream LLMs with other components of the framework. AutoPCF has shown promising results, achieving an average error rate of 42% against expert estimations and reducing PCF modeling time from days to minutes. Furthermore, other frameworks employing moderate-scale language models, such as those from the BERT family, have reached human-level accuracy in impact factor matching (Balaji et al., 2023), supply chain emissions estimation (Jain et al., 2023b), and scope 3 emissions estimations (Jain et al., 2023a) – highlighting the importance of collaborations between data scientists, software engineers, and environmental scientists. As LLMs become fine-tuned to focus on LCA tasks and extensive datasets are developed to evaluate LLM performance, LLM frameworks will become more powerful and yield greater efficiencies for completing subsets of LCA tasks, or potentially LCAs as a whole.

Yet, the complexity inherent in executing comprehensive LCAs poses a significant challenge to these frameworks as LLMs currently lack the requisite domain expertise. To navigate this complexity, future frameworks could incorporate human-in-the-loop feedback mechanisms. These mechanisms would allow for the validation and refinement of decisions based on the LLM’s reasoning and the availability of pertinent data. Moreover, enhancing reliability through iterative LLM prompting and conducting uncertainty analyses are critical steps forward as highlighted in AutoPCF (Luo et al., 2023a). Such measures would alleviate error propagation and address uncertainty challenges by delineating variations in estimations across different runs and quantifying the uncertainty in each estimation. Still, the preliminary successes of current frameworks underscore the vast potential of LLMs, particularly when integrated with advanced strategies, including prompt engineering, relevant VDBs, LCA-specific LLM fine-tuning, and automated LCA calculation methodologies. Engaging domain experts and fostering community-wide collaboration in the LCA field are pivotal towards yielding more precise and time-efficient LLM-augmented LCA frameworks.

5. Risks of using large language models for life cycle assessment tasks

There are five main risks for using LLMs for LCA tasks, including a lack of transparency, the inclusion of misinformation and bias, the limitations of the LLM due to the training data, the inability of the LLM to take responsibility for the generated content, and the lack of sustainable training processes for LLMs. Some of these risks are present in generated output and can be mitigated through prompt engineering, while others are more structural to the nature of LLMs, as shown in Fig. 5. While these risks may be mitigated through technological advancements, in the short term, LCA practitioners should be aware of these so as to use LLMs in a safe and responsible manner.

5.1. Transparency of generated content

When using LLMs, it is crucial to ensure that the content generated is transparently produced and utilized. For transparency in LLM-generated content, it is essential that the LLMs possess some level of explainability to allow humans to understand and trust the output (Barredo Arrieta et al., 2020). Due to the complexity of LLMs, the influence of specific parameter and hyperparameter values on the content generated is often not well understood. This limits the applicability of traditional ML methods (Linardatos et al., 2021) used to explain and interpret the output of LLMs. While there are methods for making an LLM more explainable, including fine-tuning methods and prompt engineering (Zhao et al., 2024), they are potentially insufficient to enable the user to trust the output from the LLM. To further increase transparency, the LCA practitioner should acknowledge the use of LLMs in conducting LCA tasks.

5.2. Misinformation and bias

Misinformation and bias in LLM output stem from two main sources: the RLHF training process and errors in the generated output. Despite RLHF’s benefits, it can reduce text diversity, hindering accessibility for diverse user groups and potentially embedding improper sustainability representations (Kirk et al., 2023). This generalization risks making LLMs less universally accessible and may lead to biased worldviews within models, affecting the quality of LCAs and their adoption by practitioners (Ferrara, 2023), (Gallegos et al., 2023). The inclusion of such biases is particularly concerning for tasks such as disseminating LCA results, where LLM outputs could influence policy decisions. LCA practitioners might mitigate these issues by critically assessing outputs, adjusting prompts, and selecting aligned LLMs. Additionally, LLM-generated text poses challenges, including repetitive or irrelevant content and "hallucinations" – outputs that conflict with inputs, lack internal consistency, or are factually incorrect (Zhang et al., 2023b), (Zhu et al., 2023b). These issues, critical throughout the LCA process, risk spreading disinformation (Zhu et al., 2023a), (Tyler et al., 2023). However, hallucinations can be readily addressed through prompt engineering, as discussed in Section 4.1.

5.3. Limitations of LLM training data

The utilization of LLMs in LCA tasks presents three main risks

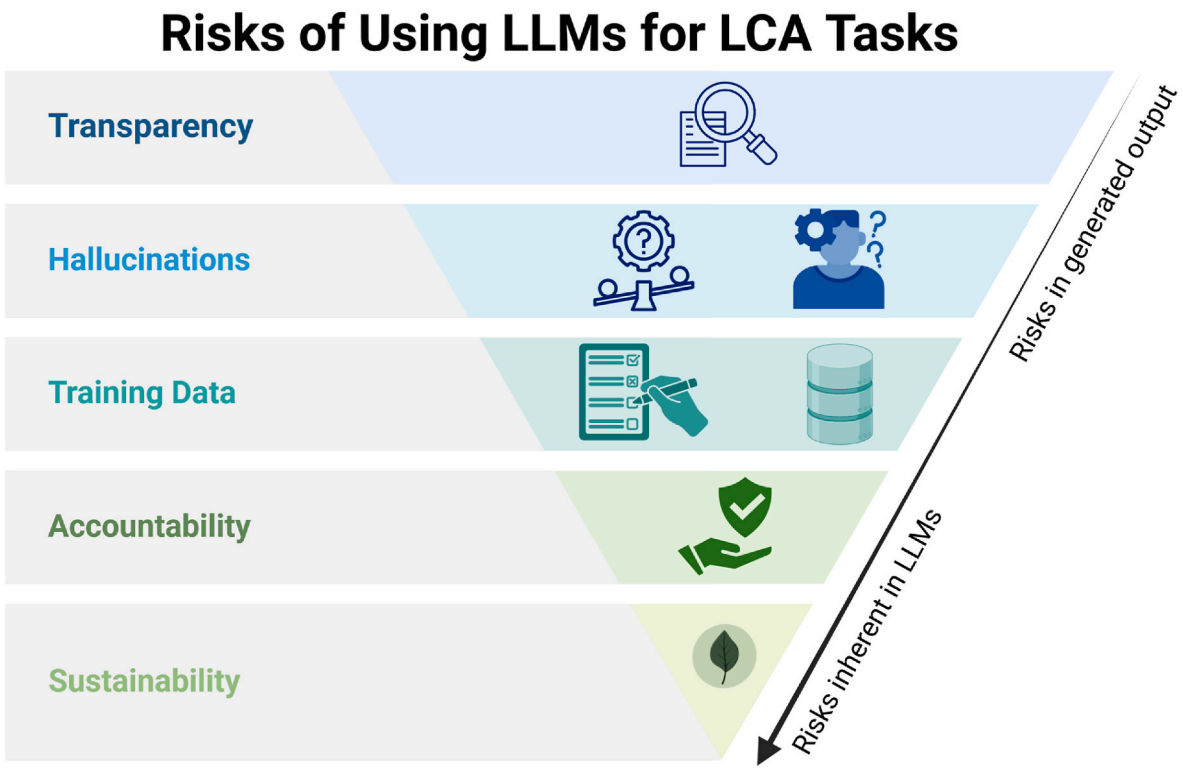


Fig. 5. The risks of using LLMs for LCA tasks, from risks present in the generated output to structural risks that are derived from the data and methods used to train the LLMs.

stemming from the collection, processing, and application of training data: sensitivity of data sources, data obsolescence, and data quality. Training LLMs on confidential or sensitive data risks privacy and security breaches (Hosseini and Horbach, 2023), especially concerning LCI data within competitive industries, as LLMs might inadvertently disclose sensitive information (Weidinger et al., 2022). Additionally, LLMs rely on static training datasets, excluding any advancements or research published after a certain cut-off date (Lazaridou et al., 2021), which may result in outdated or misleading outputs particularly detrimental for state-of-the-art LCA practices. This risk may be mitigated by integrating LLMs with continuously updated VDBs as mentioned in Section 4.2. Lastly, the inclusion of low-quality or "greenwashed" (de Freitas Netto et al., 2020) data in training sets can perpetuate misinformation, undermining the credibility of LCA results and potentially impeding climate action (Mashamaite, 2023). Thus, it is crucial for LCA practitioners to opt for LLMs trained on high-quality, recent, and aligned climate literature to ensure the effectiveness and integrity of LCA outcomes.

5.4. Accountability and responsibility

While the ethics of including LLMs in scientific research is an ongoing discussion and likely to evolve in the near future, it is currently being argued that LLMs do not have free will and therefore cannot be morally or legally held responsible for what they say (Hosseini et al., 2023). For now, LLMs are considered sophisticated tools, implying that humans will take the responsibility for any LLM-generated content (Birhane et al., 2023). Because the norms around use of LLMs in scientific research are in flux, we encourage the LCA community to collaborate with data scientists, LLM developers, and policymakers to develop standards for ethical use of LLMs in LCAs. In the interim, when an LLM has produced an output in response to a prompt, the LCA practitioner should verify and validate the output and take responsibility for decisions made based on the generated content. Moreover, the LCA practitioner has the responsibility to use LLMs in a manner consistent with local rules and regulations.

5.5. Sustainability of large language models

Due to the extensive size of LLMs and their associated training sets, there is a considerable computational cost with corresponding environmental impacts (Jiang et al., 2024). This cost applies not only to training the model but also to using it for content generation. Various metrics have been built to estimate the energy (Eilam et al., 2023), carbon (Chien et al., 2023), (Moro et al., 2023), and water (Zuccon et al., 2023) consumption of training and using LLMs, but given the difficulties of implementing these metrics as regulations (Hacker, 2023), (Wörsdörfer, 2023), these metrics will only serve to create best practices for developing LLMs. Therefore, it is recommended to first use pre-trained models and fine-tune them for highly specialized tasks; a new LLM could be developed if there is no other alternative exists (Hacker, 2023). Since most LCA tasks mentioned in Section 3 do not necessarily require a specialized LLM, using the prompt-engineering techniques discussed in Section 4.1 is likely the most sustainable approach.

6. Conclusions

Given the complex nature of the various sustainability issues, more sophisticated tools may need to be used to be able to holistically address these problems. LLMs, a class of AI models that can be used to complete a variety of natural language processing tasks, offer promise for integration into LCA studies and can accelerate progress towards achieving SDG 11, 12, and 13. LLMs offer greater time efficiency for completing LCA tasks, can provide expert-level LCA advice, increase the accessibility of LCAs, and aid in LCI data retrieval and processing tasks. These benefits are limited by the challenges of using LLMs for LCA tasks, including a lack of metrics and benchmarks to quantitatively measure the quality of the LCA, LLM data privacy issues, the difficulties experienced by LLMs

when performing calculations, and large variations in quality of the response. Moreover, there are risks in using LLMs for LCA tasks that must be acknowledged, including accountability for generated output, the inclusion of hallucination and bias in generated output, and a lack of transparency in how the LLM generates content. Because LLMs show promise, further study of their integration with LCAs is warranted, so long as the challenges and risks of using LLMs are acknowledged and mitigated. We recommend that the LCA community collaborate on ethical standards for use and best practices to ensure safe and reliable adoption of LLMs.

Because little research has so far been conducted on integrating LLMs into LCAs, this paper focuses primarily on how LLMs could be used, given their capabilities as documented in the AI literature. To this end, we have identified four primary opportunities for using LLMs to complete LCA tasks: 1) using prompt engineering to improve the quality of LLM responses, 2) integrating VDBs with LLMs to enhance LCA context awareness, data acquisition, processing, and management, benefiting not only LCI data collection but all LCA tasks, 3) the development of an LLM that specializes in LCA tasks through the use of extensive datasets based on LCA literature, and 4) integrating LLMs alongside other ML models and products into an LLM-enhanced LCA framework. To the extent that LLMs are found to be useful and beneficial, a broader conversation over whether these tools should be used in LCA studies needs to be had to set norms and guidelines for LLM use in the LCA community.

CRediT authorship contribution statement

Nathan Preuss: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Abdulelah S. Alshehri:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Fengqi You:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2024.142824.

Appendix

In the supplemental information, an example LCA scenario is provided considering the impacts of agricultural land use change in a rural community. LLMs were prompted for three reasoning tasks, including providing a goal and scope definition, functional unit, and life cycle impact methodologies. The key differences between LLM outputs in the example LCA scenario in the supplemental information are given in Table A1. A look at the presence of hallucinations in output, alongside the honesty, helpfulness, and harmlessness of the LLMs are given in Table A2. A more detailed analysis of the generated output can be found in the supplemental information.

Table A1
Summary of responses by the three LLMs to LCA reasoning prompts

Prompt	Quality	Gemini	Gemini Advanced	GPT-4
Goal and scope definition	Goal	To provide decision-makers in the small community with robust and objective information on the environmental and socioeconomic impacts of three potential land-use options	Minimize greenhouse gas emissions while considering the economic and agricultural needs of the community	To evaluate and compare the environmental, economic, and social impacts of converting agricultural land into a solar farm, maintaining it for agricultural use, or integrating both through agrivoltaics
	Goal justification	None	None	This assessment aims to inform the community's decision-making process by identifying the most sustainable option for land use that balances greenhouse gas (GHG) emission reductions, economic benefits, and social well-being.
	LCA type	Social	Not specified	Environmental, economic, and social; allocation
	LCA type justification	Consider social and environmental factors	N/A	Consider greenhouse gas emissions, biodiversity, land use change,
	Functional unit	None	1 kWh of electricity	1 GWh electricity over lifetime, maintenance of 1 ha of land use over lifetime
	Functional unit justification	None	Fair comparison of environmental impacts of different pathways	None
	Impact categories	Climate change, land use and biodiversity, water use, air and soil pollution, socioeconomic impacts	Climate change, resource depletion, job creation, tax revenue, agricultural productivity	Greenhouse gas emissions, biodiversity, land use change, upfront costs, operational costs, job creation, tax revenue, local employment opportunities, community well-being, potential conflicts or benefits to local stakeholders

(*continued on next page*)

Table A1 (*continued*)

Prompt	Quality	Gemini	Gemini Advanced	GPT-4
	Impact category justification	Impact categories address diverse perspectives	On a per category basis	None
	System boundary	Cradle-to-grave	Cradle-to-grave	Cradle-to-grave
	System boundary justification	Gives a holistic understanding of trade-offs between options	None	None
Functional unit	Functional unit	1. kWh over lifetime 2. Hectares of land used over lifetime. 3. kWh of electricity + kg of agricultural products per hectare per year over lifetime 4. Separate LCAs with separate FU for each pathway	1. 1 kWh of electricity generated. 2. 1 year of electricity provision for a typical household 3. 1 ha of land over lifetime	1. 1 GWh of electricity production 2. 1 ton of agricultural products 3. Maintaining one hectare of agricultural land use for 1 year
	Functional unit justification	1. Measures primary function of solar and agrivoltaics 2. Comparison of land use impacts 3. Capture both agriculture and energy functions. 4. Flexible weighting based on community priorities	1. Straightforward and easy to understand. 2. Contextualization of LCA results 3. Impacts related to land use change and agricultural considerations	1. Standard unit for measuring electricity production. 2. None 3. Generalized comparison for land use efficiency rather than specific outputs
Life cycle impact methodology	Choice Justification	Either Depends on the specific needs and contexts	Either Conduct assessment using both methodologies	Either For a holistic view of all potential environmental impacts, ReCiPe 2016, if focus is on specific environmental impacts for agriculture and energy, CML.

Table A2
Summary of the hallucinations contained in the generated output, and the honesty, helpfulness, and harmlessness of the generated text.

Prompt	Quality	Gemini	Gemini Advanced	GPT-4
Goal and scope definition	Hallucinations	None	1 kWh of electricity does not fairly measure impact from agricultural pathway	Conflicting life cycle impact categories
	Honesty	Yes	No	No
	Helpfulness	Yes	Speaks authoritatively	Speaks authoritatively
	Harmlessness	Yes	Yes	Yes
Functional unit	Hallucinations	None	None	None
	Honesty	Yes	Yes	Yes
	Helpfulness	Yes	Yes	Yes
	Harmlessness	Yes	Yes	Yes
Life cycle impact methodology	Hallucinations	Neither ReCiPe 2016 nor CML are good for social LCA	None	None
	Honesty	No	Yes	Yes
	Helpfulness	Yes	Yes	Speaks authoritatively
	Harmlessness	Yes	Yes	Yes

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